

Decision Structures

Section 2

Chapter 3

An Introduction to Programming Using Python
David I. Schneider

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Conditional Branching

- Topics
 - Basic *if-else* structure
 - Nested *if-else*
 - The *elif* clause
 - Input validation

if-else Statements

- Also known as branching structures
- Evaluate one or more *conditions*
- based on the result:
 - decide which **block** of code should be executed next.
 - ***block***: a sequence of one or more statements
 - **Use indentation** for appropriate block structure

```
if condition:  
    indented block of statements  
else:  
    indented block of statements
```

if-else Statements

- Example 1: Program finds larger of two numbers input by user.

```
## Determine the larger of two numbers.  
# Obtain the two numbers from the user.  
num1 = eval(input("Enter the first number: "))  
num2 = eval(input("Enter the second number: "))  
# Determine and display the larger value.  
if num1 > num2:  
    largerValue = num1 # execute this statement if the condition is true  
else:  
    largerValue = num2 # execute this statement if the condition is false  
print("The larger value is", str(largerValue) + ".")
```

[Run]

```
Enter the first number: 3  
Enter the second number: 7  
The larger value is 7.
```

if-else Statements

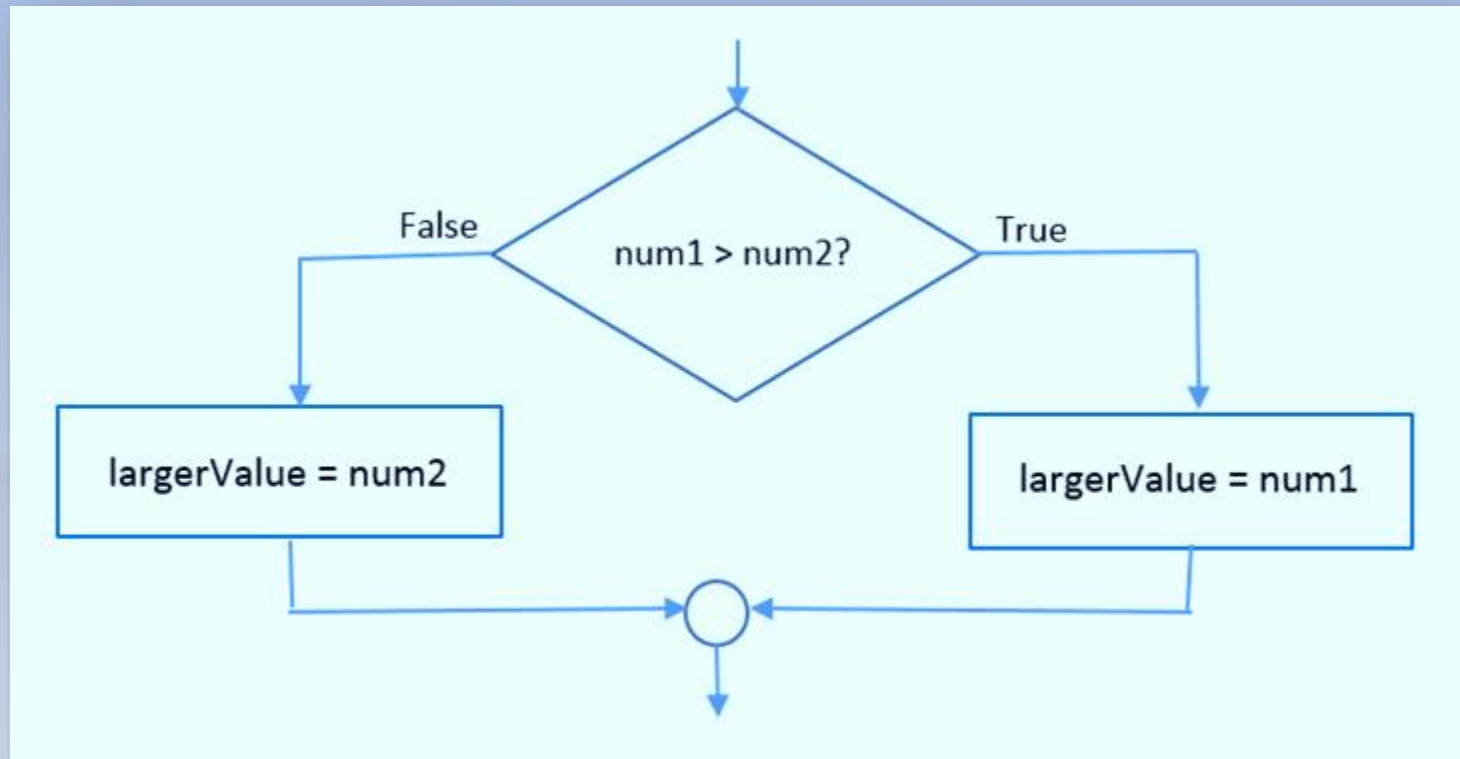


FIGURE 3.1 Flowchart for the if-else statement in Example 1.

if Statements

- The *else* part of an *if-else* statement can be omitted
- When the condition is false
 - Execution continues with line after the *if* statement block

IF Statement

- Example 1:

if $x \geq 0$:

print(x)

print('x is a positive number.')

else:

print('x is not a positive number.')

- Example 2 (else part is omitted):

if $(x > 0)$ and $(x \% 2 == 0)$:

print(x)

print('x is a positive, even number.')

Nested *if-else* Statements

- Indented blocks of *if-else* and *if* statements can contain other *if-else* and *if* statements
 - The *if-else* statements are said to be nested
 - must be indented further to indicate which *if* statement they are under
 - There may be multiple levels of nesting
 - Amount of indentation indicates which if-else statement a block of code belongs to

Nested *if-else* Statements

- Consider the task of interpreting a beacon
 - The color of the beacon light atop Boston's old John Hancock building forecasts the weather

Steady blue, clear view.
Flashing blue, clouds due.
Steady red, rain ahead.
Flashing red, snow instead.

- The following program requests a color (Blue or Red) and a mode (Steady or Flashing) as input
 - Then displays the weather forecast

Nested *if-else* Statements

```
## Interpret weather beacon.
# Obtain color and mode.
color = input("Enter a color (BLUE or RED): ")
mode = input("Enter a mode (STEADY or FLASHING): ")
color = color.upper()
mode = mode.upper()
# Analyze responses and display weather forecast.
result = ""
if color == "BLUE":
    if mode == "STEADY":
        result = "Clear View."
    else: # mode is FLASHING
        result = "Clouds Due."
else: # color is RED
    if mode == "STEADY":
        result = "Rain Ahead."
    else: # mode is FLASHING
        result = "Snow Ahead."
print("The weather forecast is", result)
```

[Run]

```
Enter the color (blue or red): RED
Enter the mode (steady or flashing): STEADY
The weather forecast is Rain Ahead.
```

Nested if-else Statement

- Example:

```
if day == 'Fri':
```

```
    if time == '10am':    # Example of a nested if statement  
        print ('Go to programming class.')
```

```
    else:
```

```
        print('Go to library')
```

```
else:
```

```
    if time == '1pm':
```

```
        print ('Go to math class.')
```

```
    else:
```

```
        print('Go home')
```

The *elif* Clause

- Allows for more than two possible alternatives with inclusion of *elif* clauses.

```
if condition1:  
    indented block of statements to execute if condition1 is true  
elif condition2:  
    indented block of statements to execute if condition2 is true  
    AND condition1 is not true  
elif condition3:  
    indented block of statements to execute if condition3 is true  
    AND both previous conditions are not true  
else:  
    indented block of statements to execute if none of the above  
    conditions are true
```

elif Clause

- Allows for more than 2 possibilities.

if age >= 60:

print('Senior admission: \$15')

elif age >= 25:

print('Adult admission: \$20')

elif age >= 12:

print('Youth admission: \$10')

else: # else part can be omitted

print('Child admission: FREE')

Input Validation

- If-else statements can be used for limited error checking.

```
int1 = int(input('Please enter a positive integer: '))  
print('You entered the number ', int1)
```

Alternatively,

```
myInput = input('Please enter a positive integer: ')  
if myInput.isdigit():  
    int1 = int(myInput)  
    print('You entered the number ', int1)  
else:  
    print('You did not enter a valid integer')
```

End

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